

Practical 9 - Estimating causal effects in trials

Survival Analysis

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Datasets required: IPASS_IPD, EXCEL_IPD, ASCOT_IPD

R packages required: survival, survRM2

```
library(survival)
library(survRM2)
```

Introduction

For this practical you will need to use 3 datasets. Each of these datasets is an approximation of data from real trials created by digitizing the Kaplan-Meier curves of each study. Datasets are stored in .csv format. Part A uses the IPASS_IPD data, Part B uses the EXCEL_IPD data and Part C uses the ASCOT_IPD data.

Part A: Questions on the IPASS data

IPASS_IPD contains data approximating the IPASS trial referenced in the lecture notes and includes the following variables:

Variable	Description
patientID	unique patient identifier
months	time from randomization in months
event	1=cancer progression or death, 0=censored
treat	1=gefitinib, 0= carboplatin plus paclitaxel
egfr	0=patient does not have EGFR mutation, 1=patient has EGFR mutation, 2= patient's EGFR status is unknown

Load the data:

```
ipass<-read.table("datasets/IPASS_IPD.csv",sep=" ",header=T)
```

Aims

Key learning objectives:

- Understand how to quantify treatment benefit using restricted mean survival time (RMST) and risk differences at specific time points.
- Understand how to assess the proportional hazards assumption and how it influences choice of summary measure.
- Encourage thinking about how the time horizon (τ) affects RMST and risk difference estimates.

Question 1

- Estimate the RMST and risk difference at 16 months using the code below.
- Write down what each means in plain English.
- Which do you prefer as a summary of treatment benefit?

```
#Kaplan-Meier by treatment
km<-survfit(Surv(months,event)~treat,data=ipass)
plot(km,ylab="Survival probability",
      xlab="Time since randomisation (months)",
      lwd=2,lty=c(1,2))
legend("topright",c("carboplatin + paclitaxel","gefitinib"),
      lwd=2,lty=c(1,2))

#RMST at 16 months
print(km,rmean=16)

#RMST (and their differences) using rmst2 function
rmst2(ipass$months, ipass$event, ipass$treat, tau=16)

#risks and risk diff at 16 months
risk.0<-1-summary(km,times=c(16))$surv[1]
risk.1<-1-summary(km,times=c(16))$surv[2]

risk.1-risk.0

#variance of risk diff, and 95% CI
varrisk.0<-summary(km,times=c(16))$std.err[1]^2
```

```

varrisk.1<-summary(km,times=c(16))$std.err[2]^2

var.riskdiff<-varrisk.1+varrisk.0

lower.ci.riskdiff<-(risk.1-risk.0)-1.96*sqrt(var.riskdiff)
upper.ci.riskdiff<-(risk.1-risk.0)+1.96*sqrt(var.riskdiff)

lower.ci.riskdiff
upper.ci.riskdiff

```

Question 2

- For EGFR mutation-positive patients only (i.e. restrict your analysis to these patients) estimate the hazard ratio comparing gefitinib to carboplatin + paclitaxel, using a Cox proportional hazards model.
- Does the proportional hazards assumption look reasonable within this subgroup of patients?

```

#cox model
cox.mod<-coxph(Surv(months,event)~treat,data=ipass[ipass$egfr==1,])
summary(cox.mod)

#take a look at Kaplan-Meier plot
km<-survfit(Surv(months,event)~treat,data=ipass[ipass$egfr==1,])
plot(km,ylab="Survival probability",
     xlab="Time since randomisation (months)",
     lwd=2,lty=c(1,2))
legend("topright",c("carboplatin + paclitaxel","gefitinib"),
     lwd=2,lty=c(1,2))

#take a look at schoenfeld residuals
sch.resid<-cox.zph(cox.mod)
plot(sch.resid)

```

Question 3

- For EGFR mutation positive patients, estimate the risk difference and RMST at 8 months. Which is the most statistically significant? Why?
- Repeat the process for patients at 16 months and comment on your findings.

```

#RMST at 8 months
print(km,rmean=8)

#RMST (and their differences) using rmst2 function
rmst2(ipass$months[ipass$egfr==1], ipass$event[ipass$egfr==1],
      ipass$treat[ipass$egfr==1], tau=8)

#risk diffs at 8m
risk.0<-1-summary(km,times=c(8))$surv[1]
risk.1<-1-summary(km,times=c(8))$surv[2]

varrisk.0<-summary(km,times=c(8))$std.err[1]^2
varrisk.1<-summary(km,times=c(8))$std.err[2]^2

var.riskdiff<-varrisk.1+varrisk.0

lower.ci.riskdiff<-(risk.1-risk.0)-1.96*sqrt(var.riskdiff)
upper.ci.riskdiff<-(risk.1-risk.0)+1.96*sqrt(var.riskdiff)

risk.1-risk.0
lower.ci.riskdiff
upper.ci.riskdiff
2*pnorm((risk.1-risk.0)/sqrt(var.riskdiff))#p-value

```

Question 4 (optional)

Discuss in groups of 3 or 4 people:

- i. Which measure of treatment effect would you recommend for a new trial focusing only on patients with an EGFR positive mutation?
- ii. After feedback from the medicine's regulator, you are told that your new trial must use restricted mean progression-free survival time as the primary measure of treatment benefit. How would you choose the milestone time for the statistical analysis?

Part B: Questions on the EXCEL data

EXCEL_IPD contains data approximating the EXCEL trial referenced in the lecture notes and includes the following variables:

Variable	Description
patientID	unique patient identifier
months	time from randomization in months
event	1=primary composite outcome occurs, 0=censored
treat	1=percutaneous coronary intervention (PCI), 0=coronary artery bypass graft (CABG)

Load the data:

```
excel<-read.table("datasets/EXCEL_IPD.csv",sep="," ,header=T)
```

Aims

Key learning objectives:

- Examine how summary measures evolve over multiple follow-up times.
- Explore stability and interpretability of RMST vs risk difference.
- Use piecewise hazards models to handle potential non-proportional hazards
- Develop critical insight into composite outcomes and how different components contribute to overall trial results.

Question 1

Estimate the RMST and risk difference at 2 years, along with 95% confidence and p-values (adapt code provided in Part A). Comment on your findings.

Question 2

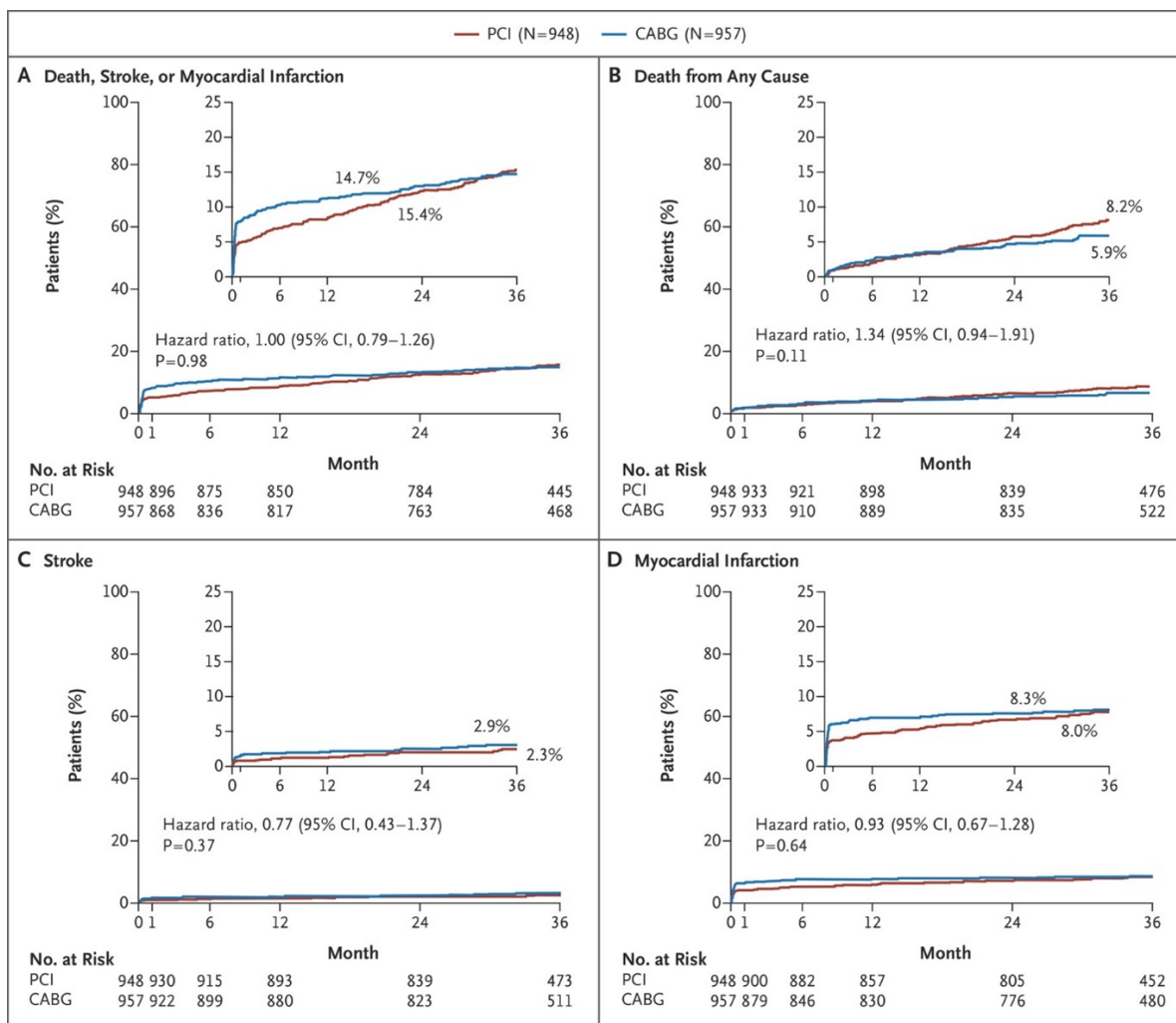
Estimate the risk difference and RMST difference at 2.5 and 3 years. What happens to the risk difference? What happens to the RMST difference?

Question 3

The figure below shows Kaplan-Meier estimates of the risks for the components of the primary outcome in the EXCEL trial are given below. What insights does this offer? Discuss in small groups how this may affect which treatment you would rather receive.

[Note that Kaplan-Meier estimates should not be used when there are competing events, as here, but that is what was presented for this study. For correct competing risks analysis see sessions 6 and 7 of the module].

```
knitr::include_graphics("09_practical_trials_fig1.png")
```



Question 4 (optional)

Estimate the hazard ratio and its 95% confidence interval within each of the following periods after randomisation: (i) the first month; (ii) 1-12 months; (iii) 12 months-36 months. We do this using piecewise hazards models.

In R the data can be split into periods using the `SurvSplit` command used in Practical 5. Look at your data to see the new structure.

```
excel.split<-survSplit(Surv(months,event)~.,cut=c(1,12),
                      start="start.time",end="stop.time",
                      event="event",episode="period",data=excel)
```

Fit separate Cox proportional hazards models within each time period using (for example):

```
cox.mod.p1<-coxph(Surv(start.time,stop.time,event)~treat,
                  data=excel.split[excel.split$period==1,])
```

What additional insight does this analysis give you? What are the limitations?

Question 5 (discuss in groups of 3 or 4)

Suppose you are an investigator planning a follow-up study to EXCEL. You wish to use the same primary composite outcome of death, stroke or myocardial infarction. Consider how you would go about statistical planning, e.g. which method for the primary analysis, when would you evaluate the outcome, which methods would you use for secondary / supportive analyses?

Part C: Questions relating to ASCOT

ASCOT_IPD contains data approximating the data for cardiovascular death (a secondary outcome) in the blood pressure lowering arm of the Anglo Scandinavian Cardiovascular Outcomes Trial (ASCOT)

Variable	Description
patientID	unique patient identifier
years	time from randomization in years
event	1=cardiovascular death occurred, 0=censored
treat	1=Amlodipine; 0=Atenolol (two blood-pressure lowering medications)

Load the data:

```
ascot<-read.table("datasets/ASCOT_IPD.csv",sep=" ",header=T)
```

Learning objectives:

- Contrast RMST vs hazard ratio to summarise the treatment effect.
- Explore how the pattern of the treatment effect over time can influence statistical power and interpretability.

Question 1

Estimate the hazard ratio, RMST and risk difference at 5 years. Comment on your findings with respect to:

- which is most meaningful?
- which is most statistically powerful and why?
- how many events are excluded for the analysis based on RMST/risk difference?
- which treatment effect is superficially most impressive and why?